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United States Patent	12416950
Kind Code	B2
Date of Patent	September 16, 2025
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Modular distributed cryogenic system

Abstract

A modular distributed cryogenic distribution system, comprising: a common chamber housing cryogenic fluid conduits; and a plurality of cryochambers connected to the common chamber. A method of operating a modular distributed cryogenic distribution system comprising a common chamber housing cryogenic fluid conduits, and plural cryochambers connected to the common chamber, includes raising one the plurality of cryochambers room temperature while a second one of the plurality of cryochambers operates at a cryogenic temperature.

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Appl. No.:	18/864314
Filed (or PCT Filed):	May 12, 2023
PCT No.:	PCT/US2023/021995
PCT Pub. No.:	WO2023/220345
PCT Pub. Date:	November 16, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20250117055 A1	Apr. 10, 2025

Related U.S. Application Data

Publication Classification**Int. Cl.:** G06F1/20 (20060101); G06N10/40 (20220101)**U.S. Cl.:****CPC** G06F1/20 (20130101); G06F2200/201 (20130101); G06N10/40 (20220101)**Field of Classification Search****USPC:** None

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Background/Summary

RELATED APPLICATIONS (1) This application claims the benefit of priority to U.S. Provisional Application No. 63/364,567 filed on May 12, 2022, the entire contents of which are incorporated by reference herein.

TECHNICAL FIELD

(1) Embodiments herein relate generally to cryogenic systems, such as cryogenic systems used for quantum computing (QC) applications, having a modular distributed architecture.

BACKGROUND

(2) A cryostat is a device that is used to maintain cryogenic temperatures (e.g., 120° K or less) for objects or materials located within the cryostat. Cryostats have been used for a number of applications in which cryogenic temperatures are desirable and/or necessary. For example, many types of quantum computing (QC) systems require quantum processing operations to be performed at extremely low temperatures. A cryostat may be used to house components of the QC system used to perform quantum processing operations such that these components may be maintained within a specified cryogenic temperature range.

SUMMARY

(3) According to one embodiment, a modular distributed cryogenic distribution system includes a common chamber housing cryogenic fluid conduits, and plural cryochambers connected to the common chamber.

(4) According to another embodiment, a method of operating a modular distributed cryogenic distribution system comprising a common chamber housing cryogenic fluid conduits, and plural cryochambers connected to the common chamber, includes raising one the plurality of cryochambers room temperature while a second one of the plurality of cryochambers operates at a cryogenic temperature.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) For a better understanding of the various described embodiments, reference should be made to the Detailed Description below, in conjunction with the following drawings in which like reference numerals refer to corresponding parts throughout the Figures.

- (2) FIG. 1 is a simplified schematic diagram illustrating an optical switch, in accordance with some example embodiments.
- (3) FIG. 2 is an illustration of a user interfacing with a hybrid quantum computing device, in accordance with some example embodiments.
- (4) FIG. 3 is schematic a top view of a modular distributed cryogenic distribution system, in accordance with some example embodiments.
- (5) FIG. 4 is a schematic top view of a network of modular distributed cryogenic distribution systems, in accordance with some example embodiments.
- (6) FIGS. 5A and 5B are rear and front perspective views, respectively, of one pod of the modular distributed cryogenic distribution system, in accordance with some example embodiments.
- (7) FIG. 6 is a side cross-sectional view of a cryochamber pod, in accordance with some example embodiments.
- (8) FIGS. 7A and 7B show a flow diagram of a method for implementing a modular distributed cryogenic system, in accordance with some example embodiments.
- (9) While the features described herein may be susceptible to various modifications and alternative forms, specific embodiments thereof are shown by way of example in the drawings and are herein described in detail. It should be understood, however, that the drawings and detailed description thereto are not intended to be limiting to the particular form disclosed, but on the contrary, the intention is to cover all modifications, equivalents and alternatives falling within the spirit and scope of the subject matter as defined by the appended claims.

DETAILED DESCRIPTION

(10) Reference will now be made in detail to embodiments, examples of which are illustrated in the accompanying drawings. In the following detailed description, numerous specific details are set forth to provide a thorough understanding of the various described embodiments. However, it will be apparent to one of ordinary skill in the art that the various described embodiments may be practiced without these specific details. In other instances, well-known methods, procedures, components, circuits, and networks have not been described in detail so as not to unnecessarily obscure aspects of the embodiments.

(11) It will also be understood that, although the terms first, second, etc. are, in some instances, used herein to describe various elements, these elements should not be limited by these terms. These terms are used only to distinguish one element from another. For example, a first electrode layer could be termed a second electrode layer, and, similarly, a second electrode layer could be termed a first electrode layer, without departing from the scope of the various described embodiments. The first electrode layer and the second electrode layer are both electrode layers, but they are not the same electrode layer.

(12) The following description, for purpose of explanation, is described with reference to specific embodiments. However, the illustrative discussions that follow are not intended to be exhaustive or to limit the scope of the claims to the precise forms disclosed. Many modifications and variations are possible in view of the above teachings. The embodiments were chosen to best explain the principles underlying the claims and their practical applications, to thereby enable others skilled in the art to best use the embodiments with various modifications as are suited to the particular uses contemplated.

(13) FIG. 1 is a simplified schematic diagram illustrating an optical switch according to an embodiment of this disclosure. Referring to FIG. 1, switch **100** includes two inputs: Input **1** and Input **2** as well as two outputs: Output **1** and Output **2**. As an example, the inputs and outputs of switch **100** can be implemented as optical waveguides operable to support single mode or multimode optical beams. As an example, switch **100** can be implemented as a Mach-Zehnder interferometer integrated with a set of 50/50 beam splitters **105** and **107**, respectively. As illustrated in FIG. 1, Input **1** and Input **2** are optically coupled to a first 50/50 beam splitter **105**, also referred to as a directional coupler, which receives light from the Input **1** or Input **2** and, through evanescent

coupling in the 50/50 beam splitter, directs 50% of the input light from Input **1** into waveguide **110** and 50% of the input light from Input **1** into waveguide **112**. Concurrently, first 50/50 beam splitter **105** directs 50% of the input light from Input **2** into waveguide **110** and 50% of the input light from Input **2** into waveguide **112**. Considering only input light from Input **1**, the input light is split evenly between waveguides **110** and **112**.

(14) Mach-Zehnder interferometer **120** includes phase adjustment section **122**. Voltage $V_{sub.0}$ can be applied across the waveguide in phase adjustment section **122** such that it can have an index of refraction in phase adjustment section **122** that is controllably varied. Because light in waveguides **110** and **112** still have a well-defined phase relationship (e.g., they may be in-phase, 180° out-of-phase, etc.) after propagation through the first 50/50 beam splitter **105**, phase adjustment in phase adjustment section **122** can introduce a predetermined phase difference between the light propagating in waveguides **130** and **132**. As will be evident to one of skill in the art, the phase relationship between the light propagating in waveguides **130** and **132** can result in output light being present at Output **1** (e.g., light beams are in-phase) or Output **2** (e.g., light beams are out of phase), thereby providing switch functionality as light is directed to Output **1** or Output **2** as a function of the voltage $V_{sub.0}$ applied at the phase adjustments section **122**. Although a single active arm is illustrated in FIG. **1**, it will be appreciated that both arms of the Mach-Zehnder interferometer can include phase adjustment sections.

(15) As illustrated in FIG. **1**, electro-optic switch technologies, in comparison to all-optical switch technologies, utilize the application of the electrical bias (e.g., $V_{sub.0}$ in FIG. **1**) across the active region of the switch to produce optical variation. The electric field and/or current that results from application of this voltage bias results in changes in one or more optical properties of the active region, such as the index of refraction or absorbance.

(16) Although a Mach-Zehnder interferometer implementation is illustrated in FIG. **1**, embodiments of this disclosure are not limited to this particular switch architecture and other phase adjustment devices are included within the scope of this disclosure, including ring resonator designs, Mach-Zehnder modulators, generalized Mach-Zehnder modulators, and the like. One of ordinary skill in the art would recognize many variations, modifications, and alternatives.

(17) In some embodiments, the optical phase shifter devices described herein may be utilized within a quantum computing system such as the hybrid quantum computing system shown in FIG. **2**. Alternatively, these optical phase shifter devices may be used in other types of optical systems. For example, other computational, communication, and/or technological systems may utilize photonic phase shifters to direct optical signals (e.g., single photons or continuous wave (CW) optical signals) within a system or network, and phase shifter architectures described herein may be used within these systems, in various embodiments.

(18) FIG. **2** is a simplified system diagram illustrating incorporation of an electro-optic switch with a prior art cryostat into a hybrid quantum computing system, according to some embodiments. In order to operate at low temperatures, for example liquid helium temperatures, embodiments of this disclosure integrate the electro-optic switches discussed herein (e.g., see FIG. **1**) into a system that includes cooling systems. Thus, embodiments of this disclosure provide an optical phase shifter that may be used within a hybrid computing system of the type illustrated in FIG. **2**. The hybrid computing system **1001** includes a user interface device **1003** that is communicatively coupled to a hybrid quantum computing (QC) sub-system **1005**. The user interface device **1003** may be any type of user interface device, for example, a terminal including a display, keyboard, mouse, touchscreen and the like. In addition, the user interface device may itself be a computer such as a personal computer (PC), laptop, tablet computer, etc.

(19) In some embodiments, the user interface device **1003** provides an interface with which a user can interact with the hybrid QC subsystem **1005**. For example, the user interface device **1003** may run software, such as a text editor, an interactive development environment (IDE), command prompt, graphical user interface, and the like so that the user can program, or otherwise interact

with, the QC subsystem to run one or more quantum algorithms. In other embodiments, the QC subsystem **1005** may be pre-programmed and the user interface device **1003** may simply be an interface where a user can initiate a quantum computation, monitor the progress, and receive results from the hybrid QC subsystem **1005**. Hybrid QC subsystem **1005** may further include a classical computing system **1007** coupled to one or more quantum computing chips **1009** (e.g., cryogenic quantum information processing chips, cryogenic single photon detector PICs). In some examples, the classical computing system **1007** and the quantum computing chip **1009** (e.g., photonic integrated circuits (PICs), electrical integrated circuits (EICs) can be coupled to other electronic components, e.g., pulsed pump lasers **1011**, microwave oscillators, power supplies, networking hardware, etc.

(20) The quantum computing chips **1009** may be housed within a cryostat, for example, cryostat **1013**. In some embodiments, each of the quantum computing chips **1009** can include one or more constituent chips, e.g., hybrid electronic chip **1015** and integrated photonics chip **1017**. The photonics chip **1017** may include the switch **100** (FIG. 1) for quantum light state preparation and detectors to perform quantum measurements (e.g., superconducting nanowire single photon detectors). Signals can be routed on- and off-chip any number of ways, e.g., via optical interconnects (e.g., optical fiber bundles) **1019** and via other electronic interconnects **1021**.

(21) Prior art large scale cryogenic distribution systems, such as the ones shown in FIG. 2, do not allow for high numbers of separate payloads to be interfaced and cooled without the entire cryogenic system warming up or changing status. This limits the flexibility of having multiple different cryogenic chambers that can interface devices together due to the different life and service cycles of these devices within the system. Such prior art large scale systems do not have modularity built into the distribution scheme for the cryogenic routing and connections and thus cannot accommodate the increased number of individual payloads. Such cryogenic systems do not allow for the independent warming and cooling and access to individual cryogenic pods because they lack individual modules for payloads.

(22) Referring to FIG. 3, a modular distributed cryogenic distribution system **300** of one embodiment (e.g., a cryogenic loop or ring system) breaks up the overall cryogenic distribution system into multiple separate cryochambers **302** (e.g., pods) attached to a common chamber **304** (e.g., loop enclosure, toroid, enclosed hoop) housing cryogenic fluid conduits **308A**, **308B**. The cryogenic fluid conduits **308A**, **308B** (e.g., pipes) are fluidly connected to the associated valves **306** and feedthrough conduits **309A**, **309B** into each cryochamber **302** for control. In one embodiment shown in FIG. 3, the common chamber **304** is an enclosure that is shaped as a toroidal loop. However, in other embodiments, the common chamber **304** may have other shapes (e.g., non-loop shapes), such as linear, curved, zig-zag, etc. The common chamber **304** can be maintained at a vacuum to provide a controlled environment to connect each of the cryochambers **302** to one another to implement distributed quantum computing tasks. In some example embodiments, the common chamber **304** comprises the coolant circulant system as well as optical interconnects (e.g., bundles of optical fiber) and electrical interconnects (e.g., cables) to connect the cryochambers through their respective interfaces to the common chamber **304**. In some example embodiments, the optical interconnects in the common chamber transfer quantum light (e.g., photonic qubits, resource states, Greenberger-Horne-Zeilinger (GHZ) entangled states) between the chambers and the common chamber **304** provides a controlled environment that protects the optical interconnects from temperature changes and physical movement, which may otherwise disrupt or decohere the quantum light on the optical interconnects.

(23) While two conduits **308A** and **308B** are shown in dashed lines as extending inside the common chamber **304** (e.g., toroidal loop enclosure), four or more conduits may be provided, as will be described below with respect to FIG. 6. For example, the conduits **308A** and **308B** may comprise liquid helium inlet and outlet conduits, respectively. If the modular distributed cryogenic distribution system **300** uses both liquid helium and nitrogen, the additional liquid nitrogen inlet

and outlet conduits may be provided inside the common chamber **304** enclosure.

(24) In one embodiment shown in FIG. **3**, the common chamber **304** may have a circular shape (e.g., when viewed from the top). However, in other embodiments, the common chamber **304** may have a polygonal (e.g., triangular, rectangular, hexagonal, etc.), an oval or an irregular shape.

(25) The embodiment shown in FIG. **3** includes a distribution scheme that is based upon a circular geometry with all the payloads (e.g., the above described chips **1009**) in separate cryochambers (e.g., “pods”) **302** arrayed around the outside of the common chamber **304**. In other embodiments, the cryochambers **302** may be arrayed inside, above and/or below the common chamber **304** in addition to or instead of outside the common chamber **304**. The center of the ring shaped loop contains a central hub **310** for connection into the larger distribution network **400** of individual modular systems (e.g., different instances of modular distributed cryogenic distribution system **300**), as shown in FIG. **4**.

(26) As shown in FIG. **4**, a main cryoplant **402** provides the cryofluid (e.g., liquid helium and/or nitrogen) into the main cryogenic supply line **404**. The main cryoplant **402** may comprise any suitable cryoplant, such as a Stirling cryoplant, pulse tube cryoplant, etc. The cryofluid is distributed from the main cryogenic supply line **404** to branch supply lines **406**. The branch lines **406** are fluidly connected to the hubs **310** of the respective modular distributed cryogenic distribution systems (e.g., the modular distributed cryogenic distribution system **300**, FIG. **3**). In some example embodiments, each branch line comprises a supply line valve **407** to fluidly isolate a set of cryogenic loops from the cryoplant **402**. In some example embodiments, the hub **310** comprises a liquid helium coolant storage tank **357A** and a liquid nitrogen coolant storage tank **357B** for storage and circulation of the different cryogenic coolants. Further, the hub **310** may contain liquefiers (e.g., a condenser/cold head to recondense the cryofluid (e.g., helium)), a vacuum pump **353** which maintain the hub **310**, the common chamber **304** and the pods at vacuum (e.g., at a pressure of about 10.sup.-6 torr), and valves. The central hub **310** is fluidly connected to the main cryogenic supply line **404** and functions as a buffer and control node for each modular distributed cryogenic distribution system. Thus, the hubs **310** provide control over the attached payloads for the given common chamber **304**. In some example embodiments, each hub **310** comprises one or more hub valves **311** to fluidly isolate the hub **310** and therefor the modular distributed cryogenic distribution system **300** from the branch line **406** and cryoplant **402**.

(27) Referring back to FIG. **3**, each common chamber **304** may be composed of several sections that combine to form the overall loop and provide an interface into the loop for the cryogenic connections and other input/outputs. Each common chamber **304** loop provides independent control over each attached payload in the respective cryochamber **302** using the valves **306**. This allows for flexibility within the larger modular distributed cryogenic distribution system **300** to warm up and cool down each modular distributed cryogenic pod on its own timeline (e.g., without warming up or cooling other pods in the same loop).

(28) The modular distributed cryogenic system **300** allows the integration of many pods into a large cryogenic supply of both liquid helium and nitrogen without the disruption of other cryochambers (e.g., cryochamber **302**) or the liquefaction of the supply. In some prior art, large scale cryogenic systems, everything is located within one larger module, and everything is cooled down at the same time with very long cooldown times due the large mass within the module. This presents problems for cryogenic systems where cycle times and overall system reliability are important across a high number of cryogenic devices. Incorporating all the cryogenic devices into one substantial module and then pressing cooldown only to have component fail results in all payloads being taken off line.

(29) In contrast, the modular distributed cryogenic distribution system **300** addresses this problem by providing an architecture that can scale up the modular distributed cryochambers (e.g., cryochamber **302**) and/or independently service or take offline individual pods to meet the overall system size required instead of just waiting on one massive chamber.

(30) Thus, each distributed cryogenic pod is just one section of a loop (e.g., ring) that can contain

more pods as the radius of the common chamber **304** increases. Each pod is a section of the common chamber **304**, and each common chamber **304** is connected to the large cryogenic supply lines through the hub **310**. The pods have control valves **306** to turn off and on the flow of the cryofluid (e.g., liquid helium and nitrogen) into the cryochamber **302** from the common chamber **304**.

(31) Depending on the overall desired system size, each modular distributed cryogenic distribution system **300** may have a desired number of pods, and the number of modular distributed cryogenic distribution systems in the network **400** can be varied to meet the level of flexibility desired for cycle time and testing. During normal full system operation, all the valves **306** are open and the devices (e.g., the chips **1009**) within the cryochambers (e.g., cryochamber **302**) are interfaced to the cooling mechanism (e.g., the main cryoplant **402**). When a fault is detected and a need arises to open one pod, the valves **306** for the specific pod close and heaters (described below) bring the cryochamber **302** (e.g., individual pod) to ambient conditions allowing for service access once vented (e.g., raised to atmospheric pressure) via pod vent **517** (e.g., to room temperature, ambient pressure of the room or environment), while the rest of the pods in the modular distributed cryogenic distribution system **300** continue to operate at cryogenic temperatures.

(32) If a larger number of pods require service or changeover, the modular distributed cryogenic distribution system **300** can be closed off from the main cryogenic supply line **404** or the branch line **406** by closing valves in the respective modular distributed cryogenic distribution system **300** and hub. A similar heating and venting process can be applied to the modular distributed cryogenic distribution system **300**, while the other systems in the network **400** continue to operate at cryogenic temperatures.

(33) The modular distributed cryogenic system **300** and network **400** of systems **300** provide an improved level of control and access compared to prior art large-scale cryogenic systems. The modular distributed cryogenic distribution system **300** is suitable for the quantum computer described above to interface many devices (e.g., chips **1009**) to a cryocooling source while maintaining a degree of flexibility for cycle time and accessibility.

(34) The cryochambers (e.g., cryochamber **302**) may contain their own set of valves **306**, heaters (described below), and feedthroughs that connect into the common chamber **304** of the modular distributed cryogenic distribution system **300** through a respective bulkhead **504** (e.g., bulkhead interface to a given pod). This allows each separate pod to go through an independent warm up and cool down cycle. Each pod may contain its own bulkhead **504** for the feedthroughs, as described below. In some example embodiments, each pod further comprises a rough pump valve **521** to couple to a vacuum pump (e.g., rough pump, manual pump) of the pod to place the pod in low or rough vacuum before the pod vacuum valve **510** (e.g., gate valve) is opened to the loop chamber. In this way, the ambient pressure in the cryochamber **302** (e.g., air from the environment in which the pod operates) is removed or mitigated and does not rush into the common chamber, and does not further rush into the other pods, which are vacuum coupled to the common chamber **304**.

(35) FIGS. 5A and 5B illustrate the rear and front views of one of the cryochambers, in accordance with some example embodiments. As shown in FIG. 5A, the back (rear) side of the cryochamber **302** which faces away from the common chamber (e.g., loop) **304** has an access door **502**. The access door **502** may be opened by service personnel to service the components inside the cryochamber **302**.

(36) As shown in FIG. 5B, the front side of the cryochamber **302** which faces the common chamber **304** contains the bulkhead **504**. The bulkhead **504** provides access from the common chamber **304** to the cryochamber **302**. The bulkhead **504** may contain various feedthroughs (e.g., openings). In other embodiments, if the cryochamber **302** is located below the common chamber **304** instead of radially outwards of the common chamber **304**, then the bulkhead **504** may be located on top of the cryochamber **302** instead of on the front side of the cryochamber **302**.

(37) As shown in FIGS. 5B and 6, the common chamber **304** may comprise a vacuum chamber which houses liquid nitrogen inlet **608A** and liquid nitrogen outlet **608B**, respectively (e.g., cryogenic fluid conduits). Liquid nitrogen inlet **608A** and liquid nitrogen outlet **608B** have corresponding feedthrough conduits **609A** and **609B** (e.g., pipes that are sealed through the bulkhead) may extend through their respective bulkhead feedthroughs **501A** and **501B** of the bulkhead **504**. The fluid feedthrough conduits **609A** and **609B** are fluidly connected through the bulkhead feedthroughs to the liquid nitrogen inlet **608A** and liquid nitrogen outlet **608B**, respectively, which are located inside the common chamber **304**.

(38) Liquid helium inlet and outlet fluid feedthrough conduits **309A** and **309B** may extend through their bulkhead respective feedthroughs bulkhead **503A** and **503B** in the bulkhead **504**. The fluid feedthrough conduits **309A** and **309B** (e.g., pipes) are fluidly connected through the feedthroughs to the liquid helium inlet and outlet conduits **308A** and **308B** (e.g., looped pipes in the ringed common chamber), respectively, which are located inside the common chamber **304**. The common chamber **304** can include the valves **306** to control circulation of the cryogenic coolants to the pods. In some example embodiments, the valves **306** comprise actuators or motors and arms **622** that extend down into the plenum to access and control valves to shut off or open respective cryogenic pipes (e.g., liquid nitrogen inlet **608A** and liquid nitrogen outlet **608B**, cryogenic fluid conduits **308A**, **308B**).

(39) The bulkhead **504** further comprises a vacuum valve **510** that can open or close the cryochamber to the pressure system of the common chamber. For example, the vacuum valve **510** can be opened and the vacuum pump of the hub **310** can pump the common chamber and any cryochambers with open vacuum valves to put the common chamber and cryochambers in a low-pressure vacuum state.

(40) In an optical bulkhead interface, optical interconnects (e.g., optical fiber bundles) **1019** extend from the common chamber **304** into the cryochamber **302** through optical feedthroughs **508** (e.g., optical ribbon cable couplers). In an electrical bulkhead interface, the electronic interconnects **1021** (e.g., direct current (DC) and/or radio frequency (RF) buses and/or wires) extend from the common chamber **304** into the cryochamber **302** through electrical feedthroughs **506**. The interconnects may pass from the common chamber **304** into the central hub **310** and then out to the remaining quantum computer components described above. In the side view of FIG. 6, the vacuum valve (e.g., gate valve) between the cryochamber **302** and the common chamber **304** is not shown, however it is appreciated that the vacuum interface between the cryochamber **302** and the common chamber **304** can include a valve to fluidly isolate the chamber from the common chamber.

(41) As shown in FIG. 6, the cryochamber **302** contains a liquid helium chamber **602**, a vacuum plenum **604** and a liquid nitrogen chamber **606**. The liquid helium chamber **602** contains liquid helium during operation of the modular distributed cryogenic distribution system **300**. The liquid helium is cycled to the liquid helium chamber **602** through conduits **308A** and **309A**, and is cycled away from the liquid helium chamber **602** through conduits **308B** and **309B**. Thus, liquid helium is routed to and from the liquid helium chamber **602** via conduits **308A** and **308B** located in the common chamber **304** and through feedthrough conduits **309A** and **309B** which extend from the common chamber **304** into the cryochamber **302** through the bulkhead **504**.

(42) The liquid helium chamber **602** may be located in the vacuum plenum **604** of the cryochamber **302**. The vacuum plenum **604** may be fluidly connected to the common chamber **304** via the feedthroughs in the bulkhead **504**. Thus, when the common chamber **304** is pumped down to a vacuum (e.g., by a vacuum pump located in the hub **310**), the vacuum plenum **604** is also pumped down to vacuum at the same time. The feedthrough conduits **309A** and **309B** may be fluidly connected to the liquid helium chamber **602** through the bulkhead **504** to cycle liquid helium to and from the chamber **602**.

(43) The quantum computing chips **1009** described above may be placed in thermal contact (such as direct or indirect physical contact) with the liquid helium chamber **602**. For example, the

quantum computing chips **1009** may be attached directly or via an interposer to the bottom of the liquid helium chamber **602**. In other embodiments, the chips **1009** may be placed on different surface(s) of the liquid helium chamber **602**, such that the liquid helium in chamber **602** cools the chips **1009** to a temperature of 4.2 K or below, such as 2-4K. Thus, the chips **1009** may be located in the vacuum plenum **604** below the liquid helium chamber **602**.

(44) The optical interconnects **1019** (e.g., optical fiber bundles) and the electronic interconnects **1021** are connected to the chips **1009** (e.g., from below and/or from the sides). The optical interconnects **1019** and the electronic interconnects **1021** extend through the vacuum plenum **604** to the respective feedthroughs in the bulkhead **504**.

(45) The liquid nitrogen chamber **606** may at least partially surround the vacuum plenum **604** and the liquid helium chamber **602**. The liquid nitrogen chamber **606** provide a thermal shield to the liquid helium chamber **602**. The fluid feedthrough conduits **609A** and **609B** cycle liquid nitrogen to and from the liquid nitrogen chamber **606**. The liquid nitrogen chamber **606** may include a nitrogen shield and contain liquid nitrogen during operation, which is maintained at a temperature of about 77K.

(46) A first heater **610** may be located adjacent to liquid helium chamber **602** to warm up the chamber **602** when the cryochamber **302** is being serviced or shut off. A second heater **612** may be located adjacent to liquid nitrogen chamber **606** to warm up the chamber **606** when the cryochamber **302** is being serviced or shut off. The heaters **610** and **612** may comprise independently controlled resistive heaters which are located at least one sidewalls of their respective chambers.

(47) The liquid helium chamber **602**, the vacuum plenum **604** and/or the liquid nitrogen chamber **606** may be attached to the cryochamber **302** using any suitable mechanical connections. For example, they may be suspended from the top of the cryochamber **302** using rods **614**.

Alternatively, they may be attached to the bottom and/or the side of the pod using one or more of rods, plates, brackets, etc.

(48) Thus, the modular distributed cryogenic distribution system **300** of one embodiment, and the network **400** of such systems **300** described above, may be used in a quantum computing device described above. The modular distributed cryogenic distribution system **300** contains separate cryopods housing photonic and electronic chips **1009** of the quantum computing device.

(49) FIGS. 7A and 7B show a flowchart of an example method **700** for implementing a modular cryogenic system, in accordance with some example embodiments. As shown in FIG. 7A, the method **700** may include providing a plurality of different cryogenic coolants to a common chamber that may include cryogenic fluid conduits to circulate the plurality of different cryogenic coolants (block **705**). For example, the hub **310** may provide a plurality of different cryogenic coolants to a common chamber **304** that may include cryogenic fluid conduits (e.g., pipes) to circulate the plurality of different cryogenic coolants (e.g., liquid helium, liquid nitrogen), as described above.

(50) As also shown in FIG. 7A, the method **700** may include providing the plurality of different cryogenic coolants from the cryogenic fluid conduits to a plurality of cryochambers (e.g., cryochamber **302**) that are connected to the common chamber **304**. In some embodiments, cryochamber houses quantum chips, such as one or more cryogenic quantum information processing chips that are cooled to a cryogenic temperature by the plurality of different cryogenic coolants (block **710**). As further shown in FIG. 7A, method **700** may include maintaining a vacuum in the common chamber and the plurality of cryochambers using a vacuum pump **353** that pumps the common chamber **304** and cryochambers with open vacuum valves (block **715**). As also shown in FIG. 7A, the method **700** may include determining, using one or more processors (e.g., classical control **1007**, terminal, desktop computer), a cryogenic quantum information processing chip in one of the cryochambers is malfunctioning (block **720**). For example, classical control **1007** may receive an error flag from a control unit that monitors the operation of the malfunctioning chip. As

further shown in FIG. 7A, method **700** may include actuating valves **306** to fluidly isolate the one of the cryochambers (block **725**). As also shown in FIG. 7A, the method **700** may include activating heaters (e.g., heaters **610**, heaters **612**) inside the one of the cryochambers to raise a temperature of the one of the cryochambers (block **730**). In some example embodiments, heating is optional and the after fluidly isolating the chamber, the chamber is vented.

(51) As further shown in FIG. 7A, the method **700** may include venting the one of the cryochambers (e.g., via vent **517**, FIG. 5B) such that the one of the cryochambers is set to ambient pressure (block **735**). As also shown in FIG. 7A, method **700** may include opening an access door **502** to the one of the cryochambers to replace the cryogenic quantum information processing chip with a replacement cryogenic information processing chip (block **740**).

(52) With reference to FIG. 7B, the method **700** may include closing the access door **502** (block **745**) to ready the chamber for cooling and vacuum pumping. As also shown in FIG. 7B, method **700** may include removing pressure from the one of the cryochambers using the vacuum pump that manages the vacuum in the common chamber (block **750**). For example, device may remove pressure from the one of the cryochambers using the vacuum pump **353** that manages the vacuum in the common chamber, as described above.

(53) As further shown in FIG. 7B, the method **700** may include actuating the valves **316** to provide the plurality of different cryogenic coolants to the one of the cryochambers to lower a temperature of the one of the cryochambers to the cryogenic temperature (block **755**).

(54) In some example embodiments, the same or similar processes of method **700** are implemented to shut down a ring of cryochambers or sets of loop cryochambers systems by opening and shutting hub valve or supply line valves, optionally heating, and venting the specific rings or pods, thereby allowing modularized access to a cryogenically cooled, electro-optical computation system in an efficient manner (e.g., without shutting down the cryoplant or the entire system). Although FIGS. 7A and 7B show example blocks of method **700**, in some implementations, method **700** may include additional blocks, fewer blocks, different blocks, or differently arranged blocks than those depicted in FIGS. 7A and 7B. Additionally, or alternatively, two or more of the blocks of method **700** may be performed in parallel.

(55) The following are example embodiments:

(56) Example 1: A modular distributed cryogenic distribution system, comprising: a common chamber housing cryogenic fluid conduits; and a plurality of cryochambers connected to the common chamber.

(57) Example 2: The modular distributed cryogenic distribution system of Example 1, wherein a first one of the plurality of cryochambers is raisable to room temperature while a second one of the plurality of cryochambers continues to operate at a cryogenic temperature.

(58) Example 3: The modular distributed cryogenic distribution system of Example 1 or Example 2, further comprising valves configured to fluidly isolate each cryochamber from the cryogenic fluid conduits in the common chamber

(59) Example 4: The modular distributed cryogenic distribution system of any one of Examples 1-3, further comprising a vacuum pump which is configured to maintain the common chamber and the plurality of cryochambers at vacuum during operation.

(60) Example 5: The modular distributed cryogenic distribution system of any one of Examples 1-4, wherein the common chamber comprises a loop, and the vacuum pump is located in a central hub chamber surrounded by the loop.

(61) Example 6: The modular distributed cryogenic distribution system of any one of Examples 1-5, wherein the plurality of cryochambers are connected to the loop via respective bulkheads comprising feedthroughs.

(62) Example 7: The modular distributed cryogenic distribution system of any one of Examples 1-6, wherein each of the plurality of cryochambers comprises a quantum computing chip.

(63) Example 8: The modular distributed cryogenic distribution system of any one of Examples 1-

7, wherein each of the plurality of cryochambers further comprises liquid helium chamber in thermal connection a quantum computing chip and fluidly connected to the cryogenic fluid conduits.

(64) Example 9: The modular distributed cryogenic distribution system of any one of Examples 1-8, wherein each of the plurality of cryochambers further comprises: a vacuum plenum enclosing the liquid helium chamber and the quantum computing chip; and a liquid nitrogen chamber at least partially enclosing the vacuum plenum and fluidly connected to the cryogenic fluid conduits.

(65) Example 10: A method comprising: providing a plurality of different cryogenic coolants to a common chamber that comprises cryogenic fluid conduits to circulate the plurality of different cryogenic coolants; providing the plurality of different cryogenic coolants from the cryogenic fluid conduits to a plurality of cryochambers that are connected to the common chamber, each cryochamber comprising a quantum computing chip that are cooled to a cryogenic temperature by the plurality of different cryogenic coolants; maintaining a vacuum in the common chamber and the plurality of cryochambers using a vacuum pump that pumps the common chamber; determining, using one or more processors, that a particular quantum computer chip in one of the cryochambers is malfunctioning; actuating valves to fluidly isolate the one of the cryochambers; venting, using a vent, the one of the cryochambers such that the one of the cryochambers is set to ambient pressure; opening an access door to the one of the cryochambers to replace the particular quantum computing chip with a replacement quantum computing chip; closing the access door; creating a vacuum in the one of the cryochambers using the vacuum pump that manages the vacuum in the common chamber; and actuating the valves to provide the plurality of different cryogenic coolants to the one of the cryochambers to lower a temperature of the one of the cryochambers to the cryogenic temperature.

(66) Example 11: The method of Example 10, wherein each cryochamber comprises a bulkhead interface to the common chamber to couple pressure between the common chamber and each cryochamber, and to further transmit data between quantum computing chips in different cryochambers of the plurality of cryochambers.

(67) Example 12: The method of Example 10 or Example 11, wherein the bulkhead interface of a cryochamber of the plurality of cryochambers comprises: a vacuum interface to control fluid coupling between the common chamber and the cryochamber; an optical signal interface to transmit optical signals between cryochambers. an electrical signal interface to transmit electrical signals between the cryochambers.

(68) Example 13: The method of any one of Examples 10-12, wherein the electrical signal interface comprises radio frequency (RF) electrical connections and direct current (DC) electrical connections; wherein the vacuum interface comprises a vacuum feed through, and wherein the vacuum interface is operable to isolate the cryochamber from the common chamber.

(69) Example 14: The method of any one of Examples 10-13, wherein the optical signal interface comprises a plurality of optical fiber couplers.

(70) Example 15: The method of any one of Examples 10-14, wherein the common chamber comprises a plurality of electrical interconnects that electrically connect to electrical signal interfaces to connect different cryochambers of the plurality of cryochambers.

(71) Example 16: The method of any one of Examples 10-15, wherein the electrical interconnects comprises electrical cables.

(72) Example 17: The method of any one of Examples 10-16, wherein the common chamber comprises a plurality of optical interconnects that optically couple to optical signal interfaces to connect different cryochambers of the plurality of cryochambers.

(73) Example 18: The method of any one of Examples 10-17, wherein the optical interconnects comprises optical fibers.

(74) Example 19: The method of any one of Examples 10-18, wherein the plurality of optical interconnects transmit photonic qubits between different cryochambers.

(75) Example 20: The method of any one of Examples 10-19, further comprising: activating heaters inside the one of the cryochambers to raise a temperature of the one of the cryochambers, wherein the heaters are activated after the valves are actuated to fluidly isolate the one of the cryochambers from the different cryogenic coolants.

(76) Example 21: The method of any one of Examples 10-20, wherein creating a vacuum in the one of the cryochambers comprises: individually pumping, using rough pump, the one of the cryochambers after the access door is closed to place the one of the cryochambers at a pressure that is nearer to a pressure of the common chamber.

(77) Example 22: The method of any one of Examples 10-21, wherein creating a vacuum in the one of the cryochambers comprises: opening a vacuum valve to fluidly couple the one of the cryochambers to the common chamber; and pumping, using the vacuum pump of the common chamber, the common chamber and each cryochamber that is fluidly coupled to the common chamber.

(78) Example 23: The method of any one of Examples 10-22, wherein the plurality of different cryogenic coolants are provided from a cryoplant.

(79) Example 24: The method of any one of Examples 10-23, wherein a cryogenic loop system comprises the common chamber and the plurality of cryochambers, wherein the cryogenic loop system further comprises a hub that includes the vacuum pump, wherein the hub is connected through a pressure bulkhead to manage pressure for the common chamber and the plurality of cryochambers.

(80) Example 25: The method of any one of Examples 10-24, wherein the hub further comprises a plurality of coolant storage tanks to receive and store the plurality of different cryogenic coolants from the cryoplant, the plurality of coolant storage tanks further connected to the cryogenic fluid conduits to provide the plurality of different cryogenic coolants to the plurality of cryochambers, wherein the hub further comprises one or more hub valves to fluidly isolate the cryogenic loop system from the cryoplant for maintenance of the cryogenic loop system without shutting down the cryoplant.

(81) Example 26: The method of any one of Examples 10-25, the cryoplant comprises a plurality of cryogenic supply lines, each cryogenic supply line providing the plurality of different cryogenic coolants to a set of cryogenic loop systems, wherein each cryogenic supply line comprises one or more supply line valves to fluidly isolate the set of cryogenic loop systems.

(82) Example 27: The method of any one of Examples 10-26, further comprising: actuating the one or more supply line valves to fluidly isolate the set of cryogenic loop systems; raising a temperature of cryochambers in the set of cryogenic loop systems using heaters in the cryochambers; venting the cryochambers using vents of respective cryochambers; and opening one or more access doors of the cryochambers.

(83) Example 28: The method of any one of Examples 10-27, wherein the cryoplant is not shut down while the replacement quantum computing chip is installed in the one of the cryochambers.

(84) Example 29: The method of any one of Examples 10-28, wherein the plurality of different cryogenic coolants comprises: liquid nitrogen and liquid helium.

(85) Example 30: The method of any one of Examples 10-29, wherein cryochamber comprises: a liquid helium chamber in thermal connection the quantum computing chip, and wherein the liquid helium chamber is fluidly connected to the cryogenic fluid conduits to circulate liquid helium in the liquid helium chamber; and wherein cryochamber further comprises: a vacuum plenum that encloses the liquid helium chamber and the one or more cryogenic quantum information processing chips; and a liquid nitrogen chamber at least partially enclosing the vacuum plenum and fluidly connected to the cryogenic fluid conduits to circulate liquid nitrogen in the liquid nitrogen chamber.

(86) The terminology used in the description of the various described embodiments herein is for the purpose of describing particular embodiments only and is not intended to be limiting. As used in

the description of the various described embodiments and the appended claims, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will also be understood that the term “and/or” as used herein refers to and encompasses any and all possible combinations of one or more of the associated listed items. It will be further understood that the terms “includes,” “including,” “comprises,” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

(87) As used herein, the term “if” is, optionally, construed to mean “when” or “upon” or “in response to determining” or “in response to detecting” or “in accordance with a determination that,” depending on the context.

(88) The foregoing description, for purpose of explanation, has been described with reference to specific embodiments. However, the illustrative discussions above are not intended to be exhaustive or to limit the scope of the claims to the precise forms disclosed. Many modifications and variations are possible in view of the above teachings. The embodiments were chosen in order to best explain the principles underlying the claims and their practical applications, to thereby enable others skilled in the art to best use the embodiments with various modifications as are suited to the particular uses contemplated.

(89) It is also understood that the examples and embodiments described herein are for illustrative purposes only and that various modifications or changes in light thereof will be suggested to persons skilled in the art and are to be included within the spirit and purview of this application and scope of the appended claims.

Claims

1. A modular distributed cryogenic distribution system, comprising: a common chamber housing cryogenic fluid conduits; a plurality of cryochambers connected to the common chamber; and valves configured to fluidly isolate each cryochamber from the cryogenic fluid conduits in the common chamber, wherein a first one of the plurality of cryochambers is raisable to room temperature while a second one of the plurality of cryochambers continues to operate at a cryogenic temperature.
2. A modular distributed cryogenic distribution system, comprising: a common chamber housing cryogenic fluid conduits; a plurality of cryochambers connected to the common chamber; and a vacuum pump which is configured to maintain the common chamber and the plurality of cryochambers at vacuum during operation.
3. The modular distributed cryogenic distribution system of claim 2, wherein the common chamber comprises a loop, and the vacuum pump is located in a central hub chamber surrounded by the loop.
4. The modular distributed cryogenic distribution system of claim 3, wherein the plurality of cryochambers are connected to the loop via respective bulkheads comprising feedthroughs.
5. A modular distributed cryogenic distribution system, comprising: a common chamber housing cryogenic fluid conduits; and a plurality of cryochambers connected to the common chamber, wherein each of the plurality of cryochambers comprises a quantum computing chip.
6. A modular distributed cryogenic distribution system, comprising: a common chamber housing cryogenic fluid conduits; and a plurality of cryochambers connected to the common chamber, wherein each of the plurality of cryochambers further comprises a liquid helium chamber in thermal connection with a quantum computing chip and fluidly connected to the cryogenic fluid conduits.
7. The modular distributed cryogenic distribution system of claim 6, wherein each of the plurality of cryochambers further comprises: a vacuum plenum enclosing the liquid helium chamber and the

quantum computing chip; and a liquid nitrogen chamber at least partially enclosing the vacuum plenum and fluidly connected to the cryogenic fluid conduits.

8. A method comprising: providing a plurality of different cryogenic coolants to a common chamber that comprises cryogenic fluid conduits to circulate the plurality of different cryogenic coolants; providing the plurality of different cryogenic coolants from the cryogenic fluid conduits to a plurality of cryochambers that are connected to the common chamber, each cryochamber comprising a quantum computing chip that are cooled to a cryogenic temperature by the plurality of different cryogenic coolants; maintaining a vacuum in the common chamber and the plurality of cryochambers using a vacuum pump that pumps the common chamber; determining, using one or more processors, that a particular quantum computer chip in one of the cryochambers is malfunctioning; actuating valves to fluidly isolate the one of the cryochambers; venting, using a vent, the one of the cryochambers such that the one of the cryochambers is set to ambient pressure; opening an access door to the one of the cryochambers to replace the particular quantum computing chip with a replacement quantum computing chip; closing the access door; creating a vacuum in the one of the cryochambers using the vacuum pump that manages the vacuum in the common chamber; and actuating the valves to provide the plurality of different cryogenic coolants to the one of the cryochambers to lower a temperature of the one of the cryochambers to the cryogenic temperature.

9. The method of claim 8, wherein each cryochamber comprises a bulkhead interface to the common chamber to couple pressure between the common chamber and each cryochamber, and to further transmit data between quantum computing chips in different cryochambers of the plurality of cryochambers.

10. The method of claim 9, wherein the bulkhead interface of a cryochamber of the plurality of cryochambers comprises: a vacuum interface to control fluid coupling between the common chamber and the cryochamber, an optical signal interface to transmit optical signals between cryochambers, and an electrical signal interface to transmit electrical signals between the cryochambers.

11. The method of claim 10, wherein the electrical signal interface comprises radio frequency (RF) electrical connections and direct current (DC) electrical connections; wherein the vacuum interface comprises a vacuum feed through, and wherein the vacuum interface is operable to isolate the cryochamber from the common chamber.

12. The method of claim 10, wherein the optical signal interface comprises a plurality of optical fiber couplers.

13. The method of claim 8, wherein the common chamber comprises a plurality of optical interconnects that optically couple to optical signal interfaces to connect different cryochambers of the plurality of cryochambers.

14. The method of claim 10, wherein the common chamber comprises a plurality of electrical interconnects that electrically connect to electrical signal interfaces to connect different cryochambers of the plurality of cryochambers.

15. The method of claim 13, wherein the optical interconnects comprises optical fibers.

16. The method of claim 14, wherein the electrical interconnects comprises electrical cables.

17. The method of claim 13, wherein the plurality of optical interconnects transmit photonic qubits between different cryochambers.

18. The method of claim 8, further comprising: activating heaters inside the one of the cryochambers to raise a temperature of the one of the cryochambers, wherein the heaters are activated after the valves are actuated to fluidly isolate the one of the cryochambers from the different cryogenic coolants.

19. The method of claim 8, wherein creating a vacuum in the one of the cryochambers comprises: individually pumping, using a rough pump, the one of the cryochambers after the access door is closed to place the one of the cryochambers at a pressure that is nearer to a pressure of the common

chamber.

20. The method of claim 19, wherein creating a vacuum in the one of the cryochambers comprises: opening a vacuum valve to fluidly couple the one of the cryochambers to the common chamber; and pumping, using the vacuum pump of the common chamber, the common chamber and each cryochamber that is fluidly coupled to the common chamber.

21. The method of claim 8, wherein the plurality of different cryogenic coolants are provided from a cryoplant.

22. The method of claim 21, wherein a cryogenic loop system comprises the common chamber and the plurality of cryochambers, wherein the cryogenic loop system further comprises a hub that includes the vacuum pump, wherein the hub is connected through a pressure bulkhead to manage pressure for the common chamber and the plurality of cryochambers.

23. The method of claim 22, wherein the hub further comprises a plurality of coolant storage tanks to receive and store the plurality of different cryogenic coolants from the cryoplant, the plurality of coolant storage tanks further connected to the cryogenic fluid conduits to provide the plurality of different cryogenic coolants to the plurality of cryochambers, wherein the hub further comprises one or more hub valves to fluidly isolate the cryogenic loop system from the cryoplant for maintenance of the cryogenic loop system without shutting down the cryoplant.

24. The method of claim 23, the cryoplant comprises a plurality of cryogenic supply lines, each cryogenic supply line providing the plurality of different cryogenic coolants to a set of cryogenic loop systems, wherein each cryogenic supply line comprises one or more supply line valves to fluidly isolate the set of cryogenic loop systems.

25. The method of claim 24, further comprising: actuating the one or more supply line valves to fluidly isolate the set of cryogenic loop systems; raising a temperature of cryochambers in the set of cryogenic loop systems using heaters in the cryochambers; venting the cryochambers using vents of respective cryochambers; and opening one or more access doors of the cryochambers.

26. The method of claim 21, wherein the cryoplant is not shut down while the replacement quantum computing chip is installed in the one of the cryochambers.

27. The method of claim 8, wherein the plurality of different cryogenic coolants comprises: liquid nitrogen and liquid helium.

28. The method of claim 8, wherein cryochamber comprises: a liquid helium chamber in thermal connection with the quantum computing chip, and wherein the liquid helium chamber is fluidly connected to the cryogenic fluid conduits to circulate liquid helium in the liquid helium chamber; and wherein cryochamber further comprises a vacuum plenum that encloses the liquid helium chamber and one or more quantum computing chips, and a liquid nitrogen chamber at least partially enclosing the vacuum plenum and fluidly connected to the cryogenic fluid conduits to circulate liquid nitrogen in the liquid nitrogen chamber.
